COMPANY PROFILE · SEPTEMBER 2026

Vital Agri Nutrients



Crop nutrition engineered, manufactured, registered and tested in Pakistan.

Lahore · since 2010 · www.van.com.pk

Vital Agri Nutrients (Pvt) Ltd

VAN formulates, manufactures, registers and tests what it sells, on 1 plant site in Lahore. Most of what leaves the plant carries a partner's name.

23
registered brands, all open to own-brand manufacturing

34
PSQCA licences: 22 on VAN brands, 12 on products ready for partners

28
crop programmes, per acre, stage by stage, free

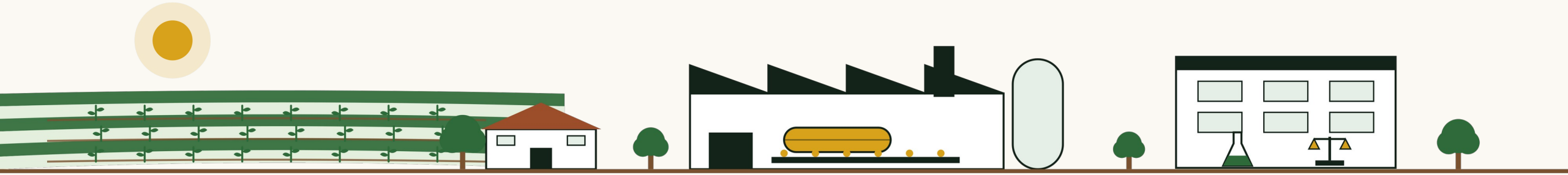
27
unbranded formulations in the library, 11 ready to carry a partner's name

100+
partner brands registered with VAN as the manufacturer, 14 partners since 2011

LAB 336
PNAC accredited, ISO/IEC 17025:2017; every batch tested before it ships

144684
Pakistan Patent, sulfur coated urea

150+
people at VAN today



HOW IT WAS BUILT

How VAN was built, 2009 to 2026.

2009 — First research farm

A year before the company. Trials every season since.

2010 — Company and land

Incorporated, and land for the plant bought.

2011 — The plant

Building started. VAN's first blended fertilizer.

2012 — Humic acid lead

The largest producer of humic acid in Pakistan.

2014 — The coated line

Boron-coated nitro potash. Soil analysis and baseline study begin.

2015 — Crop and soil

Crop-specific, soil-specific nutrition adopted.

2017 — VAN's own shops

Vital Agri Centers rolled out.

2019 — Vital Urea

Sulfur-coated urea launched. Vital Green founded.

2023 — Potash from ash

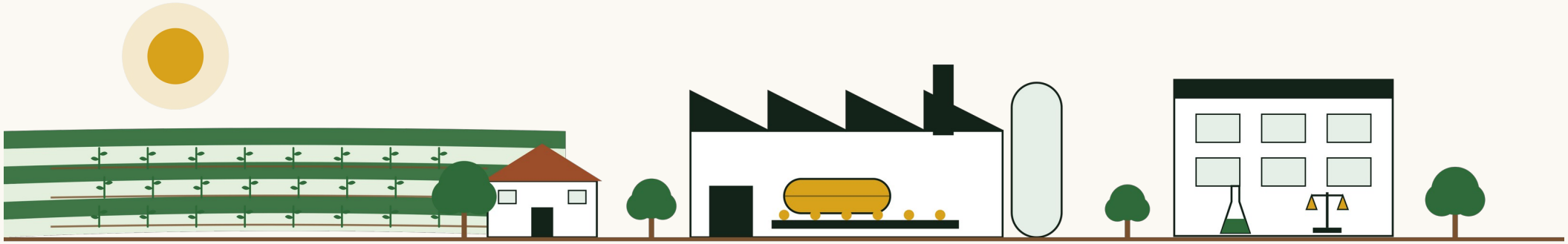
First commercial batch from crop residue ash.

2024 — Lab accredited

PNAC LAB 336, ISO/IEC 17025:2017.

2026 — Patent and bio lab

Patent 144684 granted. Biological laboratory. NP line planned.



THE PLANT

5 production lines and 2 laboratories, on 1 site.

1 Granular

Single-nutrient and specialty granules blended to a target ratio. Compatibility is checked before the batch.

2 Coated

Sulfur coating at commercial scale, the line behind Vital Urea, and boron and iron coated onto the granule. Added in 2014.

3 Liquid

The VL liquid range, from foliar concentrates to fertigation grades.

4 Water-soluble

Fully soluble grades for fertigation and foliar feeding.

5 Compounded

Every granule carries the full declared analysis, so nothing separates between the bag and the field.

• Biological laboratory

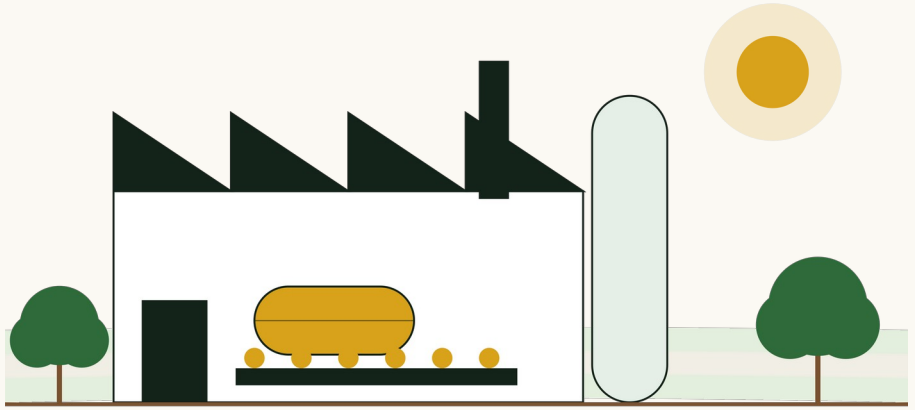
Opened in 2026. Phosphorus-solubilizing bacteria in field trials, on locally isolated strains.

• Analytical laboratory

PNAC LAB 336. Every lot is tested here before it ships.

• Packs and storage

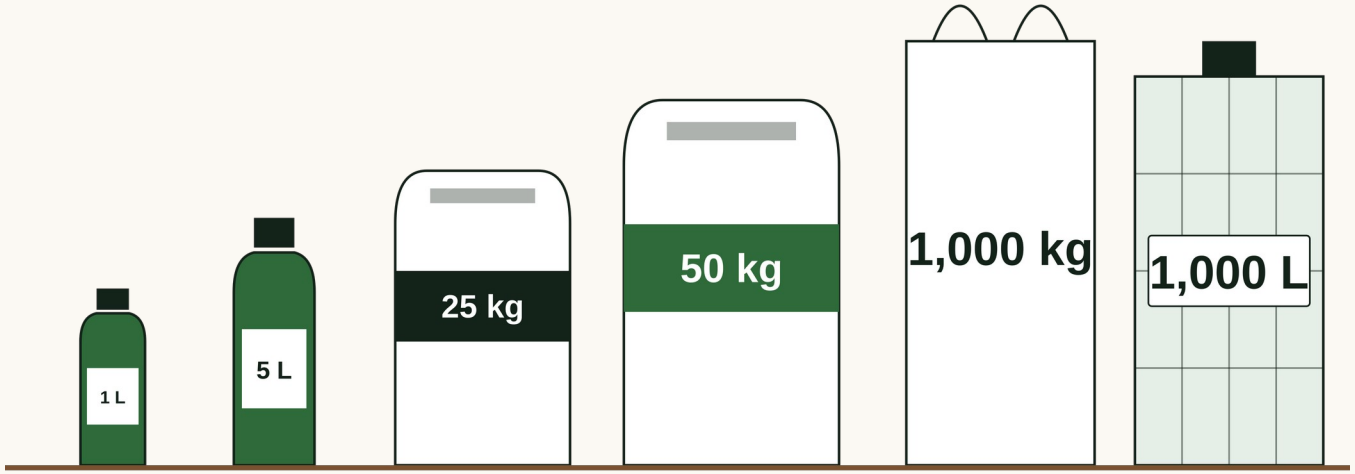
Packs from 1 litre to bulk, with finished-goods storage on site.



The analytical laboratory is on the same site as the 5 lines. Every batch from every line is tested there before it ships.

THE PLANT

7 physical forms, in packs from a 1 L bottle to a 1,000 kg bulk bag.



Physical forms

Granular, powder, crystalline soluble, liquid, pellet, water-dispersible granule and coated prill. The same composition can be made in more than 1 form.

Pack range

From a 1 L bottle to a 1,000 kg bulk bag and a 1,000 L IBC. Estate and contract-farming packs run on the same line as the retail unit.

1 t
is where a batch starts

7
physical forms, from powder to coated prill

1,000 L
the largest liquid pack, an IBC

None
no minimum on the order itself

DOOR 1 OF 4

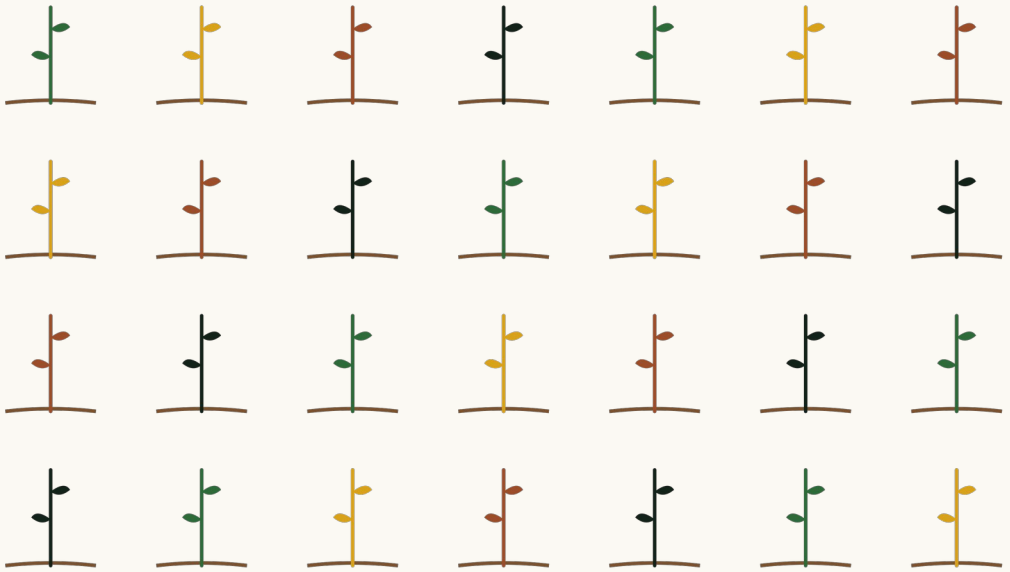
I grow



28 crop programmes, per acre, stage by stage. Your soil, your sowing date, your plan. The programmes, the soil record and the tools on van.com.pk are free, and every figure on them carries its source.

- Crop plans van.com.pk/crops
- Soil Atlas van.com.pk/soil.html
- The 17 nutrients and the barrel van.com.pk/knowledge/nutrients.html

28 crop programmes, free, per acre and stage by stage.



Wheat · Maize · Basmati rice · Hybrid rice · Cotton · Sugarcane, February planting · Ratoon sugarcane · Canola · Sesame · Soybean · Sunflower · Chickpea · Lentil · Mungbean and mash · Potato · Tomato · Chili · Garlic · Watermelon · Turmeric · Onion · Banana, year 1 · Banana, year 2 · Citrus · Date palm · Mango · Strawberry · Guava

Vitalytics, the tools

A nutrition creator on every crop page, and a live calculator on 24 crops: set the acres and the plan scales, adjusted for a soil report or the district.

Soil Atlas

The 770,160-sample Punjab record, district by district, read against what each crop removes.

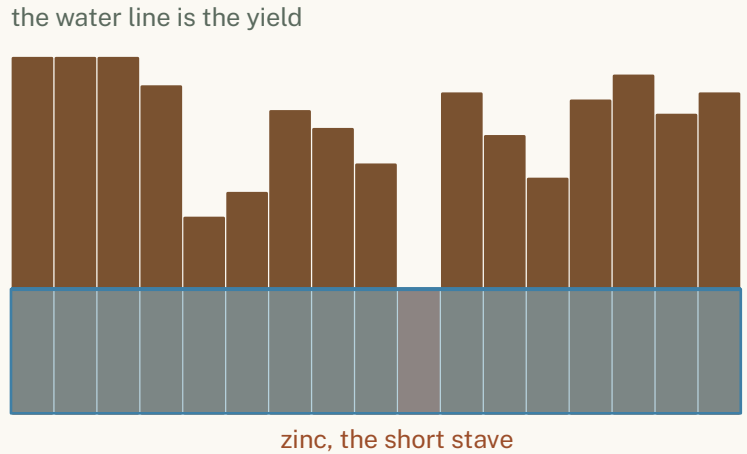
Yield and Discipline Simulator, beta

Nutrition decides about 1/4 of the yield. Sowing, seed, water, weeds and pests decide the rest. The simulator scores them on wheat and potato; 26 crops follow.

17 years of trial data from VAN's research farms sit behind every programme.

A crop grows to its shortest nutrient, not its tallest.

Liebig's Law of the Minimum, 1840s: a crop grows to the limit of the nutrient it has least of, measured against what it needs, and adding more of anything else does nothing. The picture is a barrel. Each stave is 1 nutrient; water fills it only to the shortest stave. Another bag of urea makes a tall stave taller. The water still leaves at the short one.



3 from air and water

Carbon, hydrogen, oxygen. Nobody buys them; more than 90% of the plant's dry weight.

3 the crop eats most

Nitrogen, phosphorus, potash. Most of the spend, and most of what goes wrong: too much of 1, too little of the other 2.

3 it eats in the middle

Calcium, magnesium, sulfur. Kilograms an acre, close to phosphorus, and the ones most often left out of a Pakistani programme.

8 it eats in grams

Iron, manganese, zinc, boron, copper, molybdenum, chlorine, nickel. A pinch an acre, and the crop cannot finish without it.

On van.com.pk the barrel is interactive: a grower picks his crop, enters the bags he put on an acre, and each stave is drawn against what VAN's published programme for that crop delivers. The shortest stave is where his season leaks. 11 of the 17 have a VAN product named against them.

Pakistan puts more nutrient on an acre than the world average, and grows less with each kilogram than it did in 2000.

63 kg

of nutrient on an acre of cropland, 2023. The world average is 47 kg.

83 : 1

nitrogen against potash, by tonne. India applies 11 : 1.

25%

of the nitrogen applied is taken up by the crop. The United States converts about 72%.

+68% · +31%

nutrient per acre up 68% since 2000; wheat yield up 31% over the same years.

The other 75% of the nitrogen leaves as ammonia gas, drains below the root zone as nitrate, or is lost to denitrification. The farmer pays for it at the depot and the crop never gets it. Sulfur, magnesium, zinc and boron barely register at national level.

Sources: Our World in Data / FAO, fertilizer use per hectare, 2023 (156 and 117 kg/ha, divided by 2.471). FAOSTAT, Fertilizers by Nutrient, 2023 (Pakistan N 3,808,775 t, K 45,735 t). Lassaletta et al. (2014), Environmental Research Letters 9:105011. World Bank AG.CON.FERT.ZS and the Pakistan Economic Survey Table 2.5 (1999-2000 to 2023-24). All charted and cited on van.com.pk, Knowledge.

What 770,160 soil samples say about Punjab.

8.22
mean pH. 96.1% of samples are above 7.5

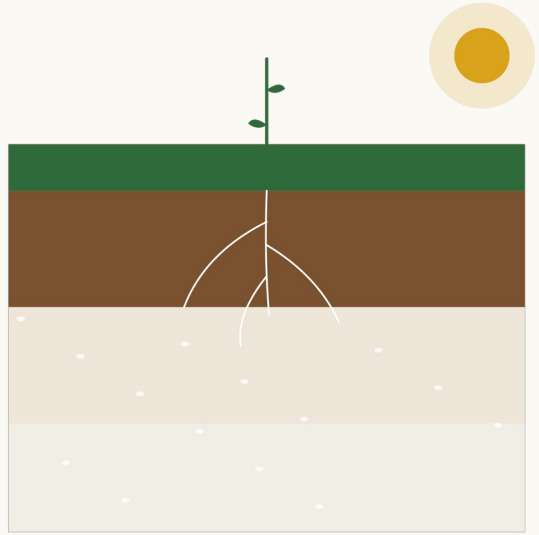
0.61%
mean organic matter. 97.6% of samples are under 1%

97.9%
of samples are below adequate in phosphorus

87.8%
are low in potassium

59.0%
are low in zinc

51.7%
are low in boron



The data is the Punjab provincial soil survey, 36 districts. VAN did not collect it. VAN reads it beside its own sampling: 18,000 records across Sindh and 600 in Gilgit-Baltistan.

0 samples tested for sulfur, in all 36 districts.

DOOR 2 OF 4

I sell



A dealership with the plant behind it. 23 brands, every batch tested before it ships, and a bag a farmer can verify by its number. The terms are on the page, in the words VAN works to.

Become a dealer van.com.pk/become-a-dealer.html

Where to buy van.com.pk/where-to-buy.html

What VAN puts behind a dealer, and the terms it works to.

The range

All 23 brands in the pack sizes the trade moves, on agreed lead times.

28 crop programmes

So the customer gets a programme, not a bag.

Brand material

VAN-branded literature and point-of-sale material.

Training and plots

Training for the team, and demonstration plots in the dealer's own territory before launch.

1 t

is where a batch starts; no minimum on the order itself

3 to 5 days

from a confirmed order to shipment for 1 to 10 t; 7 to 18 days for 11 to 100 t

15 days

from account opening to the first consignment

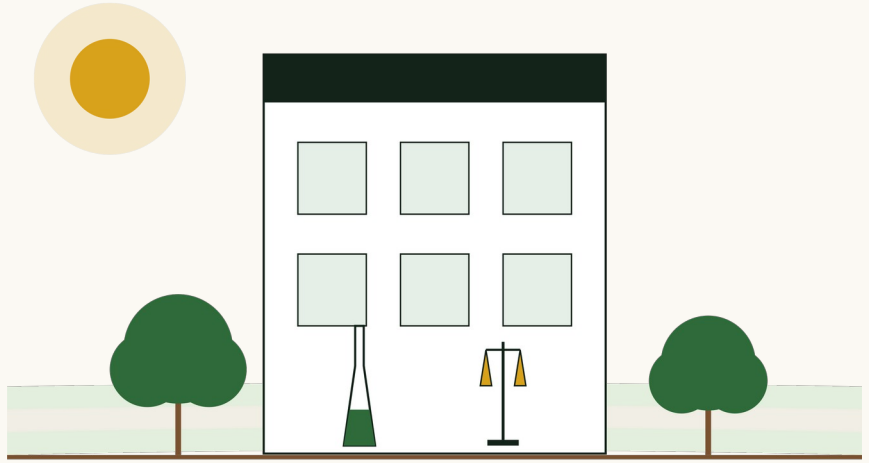
No credit

and a cash discount for advance booking, which is the other side of it

Territory terms and margin are settled at the commercial stage, per portfolio and volume. A dealer who has watched the crop respond on his own demonstration plot sells differently from one who has read a leaflet.

I SELL · QUALITY

Every batch is tested before it ships.



LAB 336

PNAC accredited, ISO/IEC 17025:2017, since 2024

8 of 21

tests accredited. All 21 are priced publicly.

2

independent analysts on every sample

3 to 5

working days from confirmed booking to report

- Written methods, calibrated instruments and reference standards run alongside every batch, so the result does not depend on who ran the test or on which day.
- A batch outside its specification is rejected.
- The certificate of analysis for any batch is free on request. Send the batch number off the bag.
- Quality management follows ISO 9001. Certification is in progress with System Certification Centre (SCC), expected during 2027.

I SELL · REGULATORY

34 PSQCA licences. The 22 on VAN brands are published by number.

34

PSQCA licences: 22 on 20 of the 23 VAN brands, 12 on products ready for partners

8

Pakistan Standards the 22 brand licences are held against

100+

partner brands registered with VAN named as the manufacturer

Every number published

Each of the 22 is listed by brand, with its licence number and its standard, on van.com.pk.

Registration run by VAN

For a partner brand, VAN files the registration under that brand, with VAN named as the manufacturer. VAN has done this more than 100 times.

Label claims checked before print

VAN checks what a label says before it is printed, and will not print a claim that a trial does not support.

DOOR 3 OF 4

Made under my name



14 partners and 100+ brands already made on this site. You bring the market. VAN does everything from the formulation to the finished bag, and files the registration under your brand with VAN named as the manufacturer.

- Your Brand van.com.pk/partner
- The 3 partnership models van.com.pk/partner#models
- How we build a range van.com.pk/partner#range

3 ways to work with VAN for the long term.

Every relationship runs 1 of these 3 ways, and a partner often runs 2 of them at once. Commercial terms are agreed per project and are not published.

Model	What VAN provides	What you provide	Who holds the registration	Time to first bag
White labelling	A formulation VAN has already developed, trialled and registered, made, tested and packed under your brand. Label and claim review before print. Repeat lots to a frozen specification.	Your brand, your artwork, your market and the selling.	VAN files it under your brand, with VAN named as the manufacturer.	About 3 months to the first commercial batch, registration included.
Joint product development	The 6 stage path from your brief: feasibility, bench formulation in the VAN lab, pilot on the commercial line, trials on VAN's research farms and on your plots where wanted, the dossier and the registration.	The brief: market, crop, target price and the constraint you are trying to beat. Your own demonstration plots where you want them.	VAN files it under your brand, with VAN named as the manufacturer.	Most briefs take 7 to 12 months. Across all 14 partners the range has been 2 to 17 months.
Toll manufacturing	Your own formulation, made on VAN's 5 lines to your specification, every lot tested in LAB 336 and released against a certificate of analysis.	The formulation, the specification, the brand and the market.	Agreed per project. VAN can file it either way.	Depends on the formulation and its registration. Agreed before the first lot.

Licensing of a VAN formulation for manufacture at another plant is not offered. The recipe and the process stay with VAN under every model.

A range, not a product.

A brand that comes with no range of its own does not have to pick from a list. VAN designs the range around the crops, the regions and the channel you sell into, from what is already made here, and then keeps adding to it.

1 — YOU

Your crops, regions and channel

Which crops your customers grow, which districts, and whether you sell through dealers, to estates or to corporate farms. That is the whole brief for a first range.

2 — VAN

A launch range from the 23 brands

VAN proposes the products a first season needs, each with the stage it goes on at and the pack it comes in, built from the same programmes published on the site.

3 — VAN

A roadmap from the library

What to add in the second and third season, from the 27 codes in the formulation library, and which of them would need a trial first.

4 — BOTH

Each product on its own route

A library formulation reaches you in about 3 months. A product developed to your brief takes a season. The range grows on that clock, not all at once.

The formulation library: 27 codes. 11 ready to carry your name, 8 validated, 3 in field trials, 3 in development, 1 at concept, 1 at feasibility.

The portfolio, without the names.

VAN does not publish its partners' names. These figures describe the portfolio without them. A partner under NDA can be given specifics in a meeting.

100+

partner brands registered with VAN named as the manufacturer

14

companies hold them between them

2010

the year the longest-running partnership started

8

products held by the largest single partner. The next largest holds 7.

27

unbranded formulations in the library, 11 ready to carry your name

2 to 17

months from brief to market, the fastest and the slowest so far

7 to 12

months for most briefs

Who already works this way: multinational crop-protection companies, national agri-input companies, sugar mills and corporate farms, and importers replacing a bought-in product.

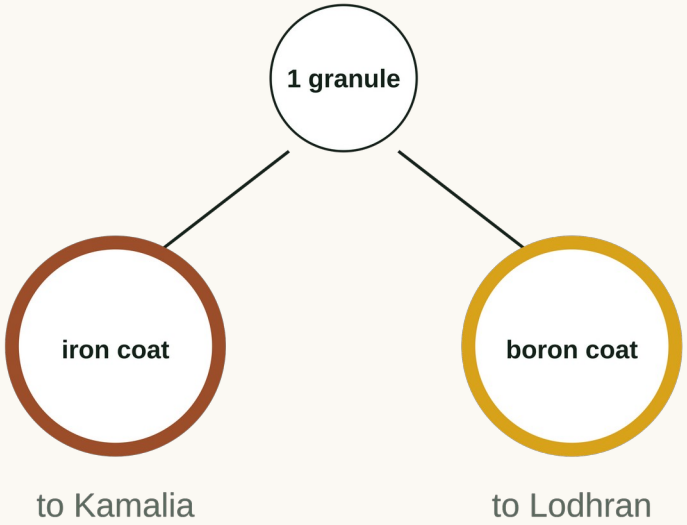
VAN designs each grade for the soil where it will be used.

770,160
soil records across Punjab

18,000
across Sindh

600
in Gilgit-Baltistan

The phosphate granule sent to Kamalia carries an iron coat. The same granule going to Lodhran carries boron. It is 1 base material on 1 coating line, and the micronutrient is chosen for the soil where the bag will be opened.



VAN does not sell it for you

No sales force, no dealers, no farmer demand. You own the market and the selling.

VAN does not hand over the formulation

You sell it under your name. The recipe and the process stay with VAN.

No claim a trial does not support

VAN will not put it on your label, even when your marketing department wants it.

DOOR 4 OF 4

I want to do it differently



For the grower who wants to test the practice against the numbers, and for the company that wants a product nobody sells yet. The 3 flagships below are VAN's own answers to 3 questions a farmer asks.

- Vitalytics, the tools van.com.pk/tools.html
- The formulation path van.com.pk/partner/pipeline.html
- The composition chart van.com.pk/composition.html

THE 3 FLAGSHIPS

The soil takes fertilizer back in 3 ways. VAN makes a product against each, and answers the farmer's question first.

Vital Urea · SCU

"I put 2 bags of urea on and the field goes yellow again in 3 weeks. Where did it go?"

A bag of urea has 3 claimants: the crop, the air and the water. The sulfur coating limits how much is exposed at once, so the crop meets the nitrogen while it is still there.

Green Phosphate · 6-32-0

"A bag of DAP every year and the soil test still says phosphorus is low. Where does it go?"

Calcareous soil above pH 8 takes the phosphate back. The granule carries its own acidic zone, sealed with humic, which slows that reaction for the weeks a root needs.

Vital Potash · 11-0-44

"I fed the crop well and it still filled light. Why?"

The crop grows on nitrogen and fills on potash. Made for the fill window, with boron in 2 forms coated onto the granule since 2014.



FLAGSHIP 1 OF 3 · VITAL UREA

Sulfur coated urea, patented and made in Pakistan: 1 bag, once, at the first application.

144684

Pakistan Patent, held by VAN. Launched in 2019.

25 kg

bag. N 32%, S 13%: at least 8.0 kg of nitrogen and 3.25 kg of sulfur in every bag.

2012

VAN began coated work and has tested many combinations since.

Broadcast

drill or side-dress it. Never fertigate.

Over days

the coating breaks down in steps once water reaches it.

Trial	Year	Run by
Rafhan Maize Products	2023	Joint, VAN with Rafhan
Descon Research Farm	2024	Joint, VAN with Descon
Independent partner farm, not named	2025	Independent

1 bag of Vital Urea (25 kg) takes the place of 1 bag of urea (50 kg) once in the crop cycle, at the first application; the rest of the nitrogen programme stays as it is. Trial summaries are with the commercial team.

What VAN will not claim: a yield figure, from the 3 maize trials or any other; a laboratory release curve with days on it; that the coating stops volatilisation; a soil pH or soil test change. Do not compare kilograms. Compare what reaches the crop in the weeks it can use it.

Do not compare kilograms. Compare what reaches the crop.

Green Phosphate · 0-32-0 · 35 and 50 kg

Calcareous soil takes the phosphate back

Above pH 8 with free lime, phosphate dissolving out of a granule meets calcium within a few centimetres and drops out of solution; the loss is its slow ageing into forms the root cannot use. 35 of 36 Punjab districts read Critical or Weak on phosphorus.

The granule carries its own acidic zone

Granulated in an acidic mix and sealed with humic, with a fraction of fulvic. The zone around each granule starts acidic, which slows the calcium reaction for the weeks a root needs to reach it. A microsite effect; it will not move a soil test.

What the humic does

It feeds the soil and the root: across 81 published studies humic raised shoot and root growth (Rose et al. 2014), and Pakistani calcareous-soil trials found the consistent gain came from organic matter applied with the phosphate.

Will not claim: that 70 to 80% of applied phosphate is fixed; that it beats DAP or TSP by source alone; a soil-test change; a replicated independent trial (none exists yet).

Vital Potash · N 11 K_2O 44 · 20 kg

The crop grows on nitrogen and fills on potash

Potash moves the sugar the leaf makes into the grain, the tuber or the fruit, and thickens the cell wall against heat, drought and disease. Pakistan applies about 1 kg of potash for every 83 kg of nitrogen.

Made for the fill window

Fully water soluble; it goes on at grand growth and maturity, through the fertigation or flood water. The small nitrogen share keeps the plant working through fill without pushing new leaf.

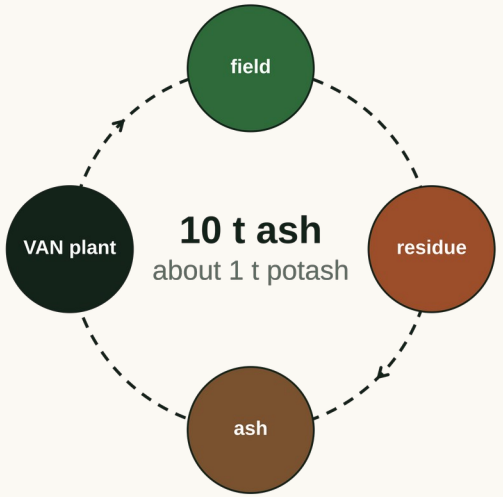
Boron rides on the granule, in 2 forms

Since 2014 VAN has coated boron onto the potash granule on its own line, developed and built locally: sodium borate for fast action and calcium borate for slow release. Named, never numbered.

Will not claim: a boron percentage; a yield, disease or immunity effect; a foliar rate; that 2 applications are the season's whole potash.

WHERE THE NUTRIENTS GO AFTER HARVEST

VAN recovers potash from crop-residue ash, in place of imported potash.



10 t

of crop residue ash gives about 1 t of potash

3,000 t

of ash processed so far, through VAN's own plant

About 20%

of VAN's potash intake now comes from this route

2027

a larger recovery line is planned

Silicon

is recovered from the same ash. In its final R&D stage; in no product yet.

V-Compost

is made from plant residue only, screened before it goes in. It adds organic matter to soil under 1% organic matter.

The minerals that left the field with the residue go back onto the land.

What VAN is developing now.

- **Potash recovered in Pakistan**

  The crop residue ash route, taken to a larger line.

- **Biologicals for nitrogen and phosphorus**

  Phosphorus-solubilizing bacteria and a seed treatment in field trials, a liquid inoculant in development. All on locally isolated strains.

- **Mycorrhizal inoculant**

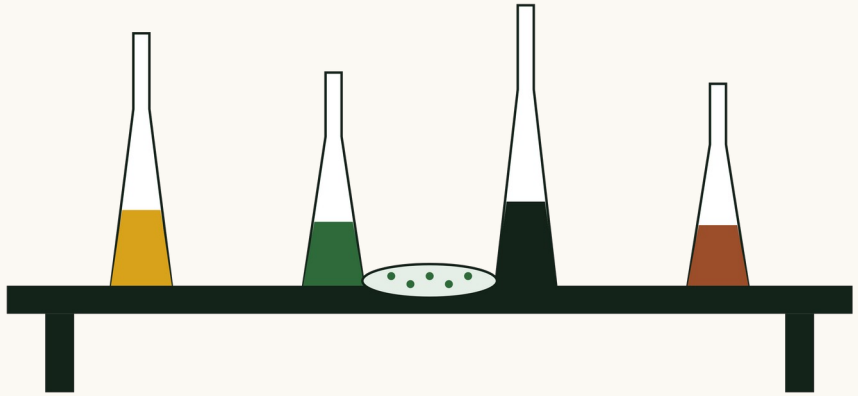
  In development. It extends the root system that has to reach the phosphorus the bacteria free up.

- **New coated and phosphate-efficiency products**

  In development: a dual coated potash at concept, a dual release phosphate at feasibility, in the published library a K-50DC and P-DRC.

- **An NP granulation line**

  Planned for the plant site. Work has not started and no date is set.



The formulation library today: 27 formulations. 11 ready to carry your name, 8 validated, 3 in field trials, 3 in development, 1 at concept, 1 at feasibility. Every code is on van.com.pk, with its published meta.

TALK TO VAN

Write to the address for what you need.

I grow

kisan@van.com.pk

A crop plan, product rates, a dealer near you.

Made under my name

partner@van.com.pk

White labelling, joint product development, toll manufacturing.

Everything else

info@van.com.pk

Supply, distribution, the laboratory, a plant visit.

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